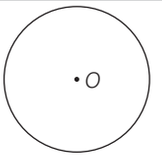
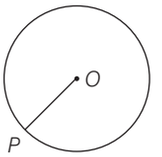
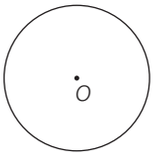
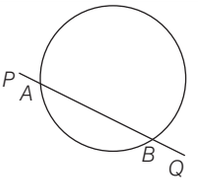
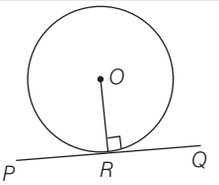
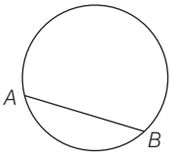
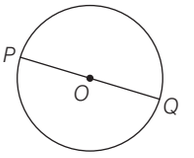
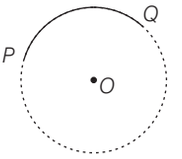
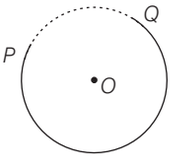
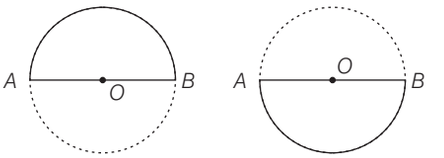
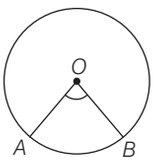
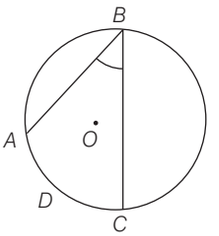
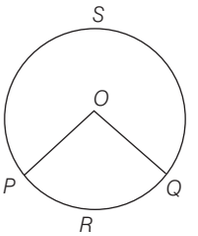
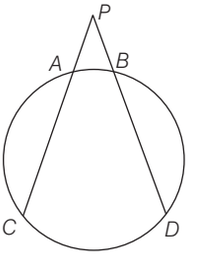
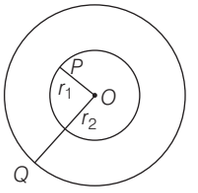
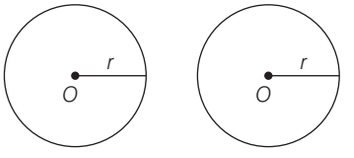
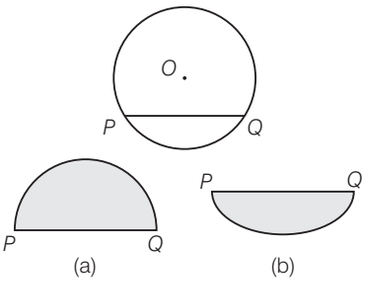
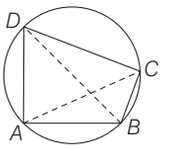
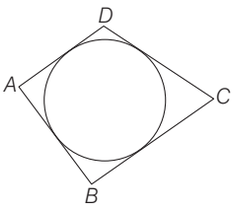
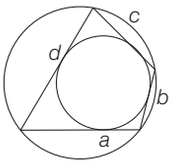
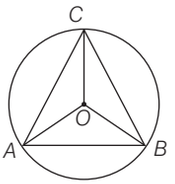
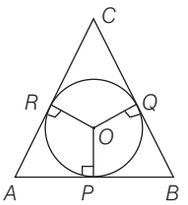
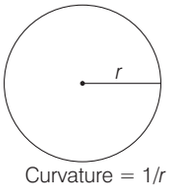
12.6 Circles

Definition : A circle is a set of points on a plane which lie at a fixed distance from a fixed point.

Nomenclature	Definition	Diagram
Centre	The fixed point is called the centre. In the given diagram '*O*' is the centre of the circle.	
Radius	The fixed distance is called a radius. In the given diagram *OP* is the radius of the circle. (Point *P* lies on the circumference)	
Circumference	The circumference of a circle is the distance around a circle, which is equal to $2\pi r$. ($r \rightarrow$ radius of the circle)	
Secant	A line segment which intersects the circle at two distinct points, is called as secant. In the given diagram secant *PQ* intersects circle at two points *A* and *B*.	
Tangent	A line segment which has one common point with the circumference of a circle *i.e.,* it touches only at one point is called as tangent of circle. The common point is called as point of contact. In the given diagram *PQ* is a tangent which touches the circle at a point *R*.	 (*R* is the point of contact) **NOTE** Radius is always perpendicular to tangent.
Chord	A line segment whose end points lie on the circle. In the given diagram *AB* is a chord.	
Diameter	A chord which passes through the centre of the circle is called the diameter of the circle. The length of the diameter is twice the length of the radius. In the given diagram *PQ* is the diameter of the circle. ($O \rightarrow$ is the centre of the circle)	
Arc	Any two points on the circle divides the circle into two parts the smaller part is called as minor arc and the larger part is called as major arc. It is denoted as '$\overset{\frown}{\quad}$'. In the given diagram $\overset{\frown}{PQ}$ is arc.	 $PQ \rightarrow$ minor arc  $PQ \rightarrow$ major arc

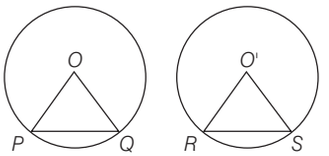
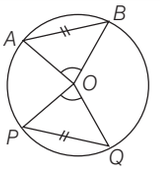
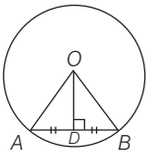
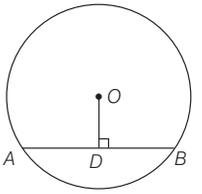
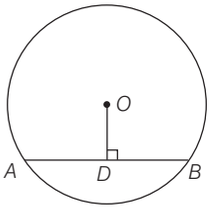
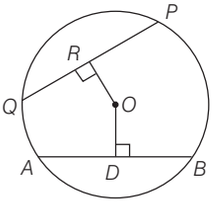
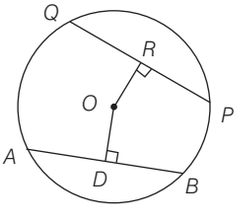
Nomenclature	Definition	Diagram
Semicircle	A diameter of the circle divides the circle into two equal parts. Each part is called as semicircle.	
Central angle	An angle formed at the centre of the circle, is called the central angle. In the given diagram $\angle AOB$ is the central angle.	
Inscribed angle	When two chords have one common end point, then the angle included between these two chords at the common point is called the inscribed angle. $\angle ABC$ is the inscribed angle by the arc $\overset{\frown}{ADC}$.	
Measure of an arc	Basically it is the central angle formed by an arc. *e.g.,* (a) measure of a circle = 360° (b) measure of a semicircle = 180° (c) measure of a minor arc = $\angle POQ$ (d) measure of a major arc = 360 − $\angle POQ$	 m (arc PRQ) = $m\angle POQ$ m (arc PSQ) = 360° − m (arc PRQ)
Intercepted arc	In the given diagram $\overset{\frown}{AB}$ and $\overset{\frown}{CD}$ are the two intercepted arcs, intercepted by $\angle CPD$. The end points of the arc must touch the arms of $\angle CPD$ *i.e., CP* and *DP*.	
Concentric circles	Circles having the same centre at a plane are called the concentric circles. In the given diagram there are two circles with radii r_1 and r_2 having the common (or same) centre. These are called as concentric circles.	
Congruent circles	Circles with equal radii are called as congruent circles.	

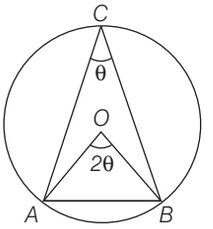
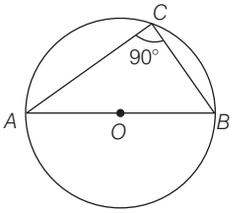
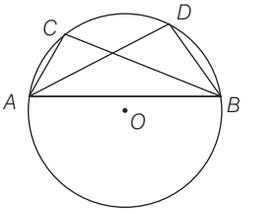
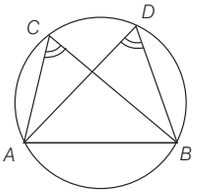
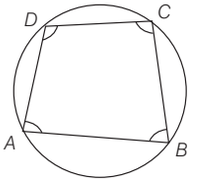
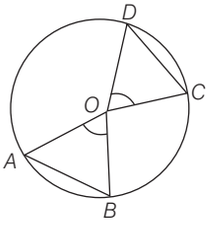
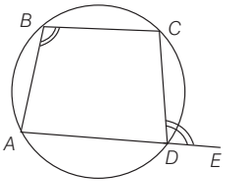
Nomenclature	Definition	Diagram
Segment of a circle	A chord divides a circle into two regions. These two regions are called the segments of a circle. (a) major segment (b) minor segment	
Cyclic Quadrilateral	A quadrilateral whose all the four vertices (A, B, C, D) lie on the circle. For a cyclic quadrilateral $ABCD$, we must have $(AB \times CD) + (BC \times AD) = (AC \times BD)$	
Tangential Quadrilateral	A tangential quadrilateral or circumscribed quadrilateral is a convex quadrilateral whose sides are all tangent to a single circle within the quadrilateral . For a tangential quadrilateral, we must have $AB + CD = BC + AD$	
Bicentric Quadrilateral	A quadrilateral is said to be bicentric if it is cyclic and tangential.	
Circumcircle	A circle which passes through all the three vertices of a triangle. Thus the circumcentre is always equidistant from the vertices of the triangle. $OA = OB = OC$ (circumradius)	
Incircle	A circle which touches all the three sides of a triangle *i.e.,* all the three sides of a triangle are tangents to the circle is called an incircle. Incircle is always equidistant from the sides of a triangle. $OP = OQ = OR$ (inradius of the circle)	
Curvature	Curvatures is a measure of how quickly a tangent line turns on a curve. We would expect the curvature to be 0 for a straight line, to be very small for curves which bend very little and to be large for curves which bend sharply. Larger the circle, smaller the curvature. Curvature of a circle $= \frac{1}{\text{radius}}$	

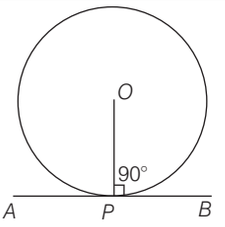
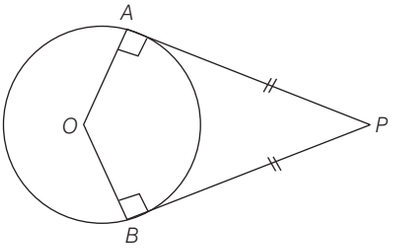
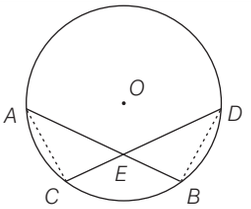
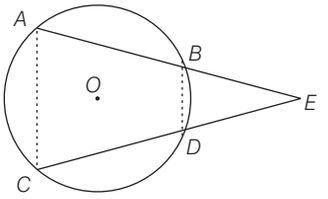
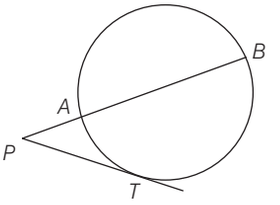
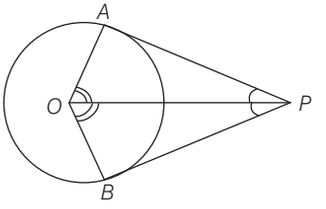
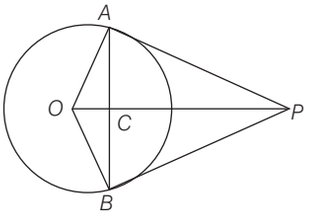
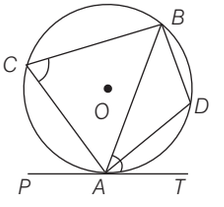
Theorems

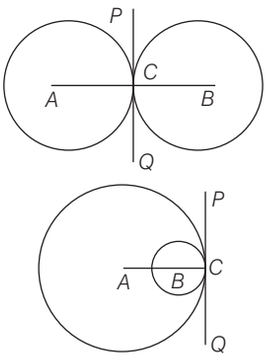
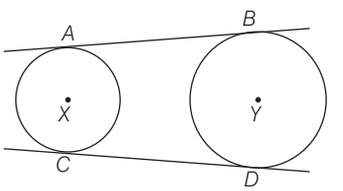
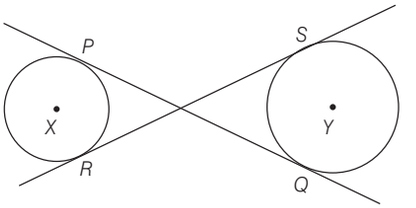
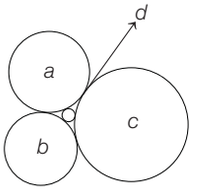
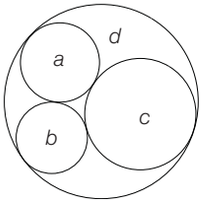
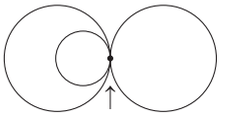
NOTE

- Two arcs of a circle (or of congruent circles) are congruent if their degree measures are equal.
- There is one and only one circle passes through three non-collinear points.

S. No.	Theorem	Diagram
1.	In a circle (or in congruent circles) equal chords are made by equal arcs. $\{OP = OQ\} = \{O'R = O'S\}$ and $PQ = RS$ $\therefore$ $PQ = RS$	
2.	Equal arcs (or chords) subtend equal angles at the centre $\because$ $PQ = AB$ $\therefore$ $\angle POQ = \angle AOB$	
3.	The perpendicular from the centre of a circle to a chord bisects the chord *i.e.,* if $OD \perp AB$ $\therefore$ $AB = 2AD = 2BD$	
4.	The line joining the centre of a circle to the mid-point of a chord is perpendicular to the chord. $\because$ $AD = DB$ $\therefore$ $OD \perp AB$	
5.	Perpendicular bisector of a chord passes through the centre. *i.e.,* if $OD \perp AB$ and $AD = DB$ $\therefore$ O is the centre of the circle.	
6.	Equal chords of a circle (or of congruent circles) are equidistant from the centre. $\because$ $AB = PQ$ $\therefore$ $OD = OR$	
7.	Chords which are equidistant from the centre in a circle (or in congruent, circles) are equal. $\because$ $OD = OR$ $\therefore$ $AB = PQ$	

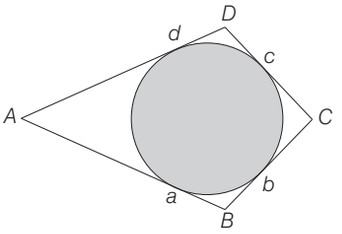
S. No.	Theorem	Diagram
8.	The angle subtended by an arc (the degree measure of the arc) at the centre of a circle is twice the angle subtended by the arc at any point on the remaining part of the circle. $m\angle AOB = 2m\angle ACB.$	
9.	Angle in a semicircle is a right angle.	
10.	Angles in the same segment of a circle are equal. *i.e.,* $\angle ACB = \angle ADB$	
11.	If a line segment joining two points subtends equal angle at two other points lying on the same side of the line containing the segment, then the four points lie on the same circle. $\angle ACB = \angle ADB$ $\therefore$ Points *A, C, D, B* are concyclic *i.e.,* lie on the circle.	
12.	The sum of pair of opposite angles of a cyclic quadrilateral is 180°. $\angle DAB + \angle BCD = 180°$ and $\angle ABC + \angle CDA = 180°$ (Inverse of this theorem is also true)	
13.	Equal chords (or equal arcs) of a circle (or congruent circles) subtended equal angles at the centre. $AB = CD$ (or $AB = CD$) $\therefore$ $\angle AOB = \angle COD$ (Inverse of this theorem is also true)	
14.	If a side of a cyclic quadrilateral is produced, then the exterior angle is equal to the interior opposite angle. $m\angle CDE = m\angle ABC$	

S. No.	Theorem	Diagram
15.	A tangent at any point of a circle is perpendicular to the radius through the point of contact. (Inverse of this theorem is also true)	
16.	The lengths of two tangents drawn from an external point to a circle are equal. *i.e.,* $AP = BP$	
17.	If two chords AB and CD of a circle, intersect inside a circle (outside the circle when produced at a point E), then $AE \times BE = CE \times DE$	 
18.	If PB be a secant which intersects the circle at A and B and PT be a tangent at T, then $PA \cdot PB = (PT)^2$	
19.	From an external point from which the tangents are drawn to the circle with centre O, then (a) they subtend equal angles at the centre. (b) they are equally inclined to the line segment joining the centre of that point. $\angle AOP = \angle BOP$ and $\angle APO = \angle BPO$	
20.	If P is an external point from which the tangents to the circle with centre O touch it at A and B, then OP is the perpendicular bisector of AB. $OP \perp AB$ and $AC = BC$	
21.	**Alternate segment theorem :** If from the point of contact of a tangent, a chord is drawn, then the angles which the chord makes with the tangent line are equal respectively to the angles formed in the corresponding alternate segments. In the adjoining diagram. $\angle BAT = \angle BCA$ and $\angle BAP = \angle BDA$	

S. No.	Theorem	Diagram
22.	The point of contact of two tangents lies on the straight line joining the two centres. (a) When two circles touch externally, then the distance between their centres is equal to sum of their radii. *i.e.,* $AB = AC + BC$ (b) When two circles touch internally, the distance between their centres is equal to the difference between their radii. *i.e.,* $AB = AC - BC$	
23.	For the two circles with centre X and Y and radii r_1 and r_2. AB and CD are two Direct Common Tangents (DCT), then the length of DCT $= \sqrt{(\text{distance between centres})^2 - (r_1 - r_2)^2}$	
24.	For the two circles with centre X and Y and radii r_1 and r_2. PQ and RS are two transverse common tangent, then length of TCT $= \sqrt{(\text{distance between centres})^2 - (r_1 + r_2)^2}$	
25.	**Descartes' circle equation theorem :** Given four mutually tangent circles with curvatures a, b, c and d, then $(a + b + c + d)^2 = 2(a^2 + b^2 + c^2 + d^2)$	
	If any one of the circles is circumscribing, say a circle of curvature d is circumscribing the circles of curvatures a, b and c, then $(a + b + c - d)^2 = 2(a^2 + b^2 + c^2 + (-d^2))$	
	In lay terms, if all the points of tangency are external, the curvatures are considered positive, but if one circle encompasses the others, that circle has negative curvature. Here, as all three circles are tangent to each other at the same point, Descartes' theorem does not apply.	

12.7 **Tangential Quadrilateral**

A tangential quadrilateral or circumscribed quadrilateral is a convex quadrilateral whose sides are all tangent to a single circle within the quadrilateral. This circle is called the incircle of the quadrilateral; its center is the incenter and its radius is called the inradius.

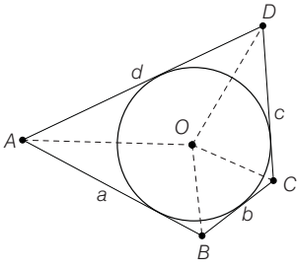


All triangles have an incircle, but not all quadrilaterals do. An example of a quadrilateral that cannot be tangential is a non-square rectangle.

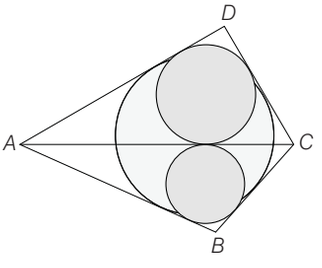
If a quadrilateral is both tangential and cyclic, it is called a bicentric quadrilateral. Examples of tangential quadrilaterals are squares, rhombi, and kites.

Properties

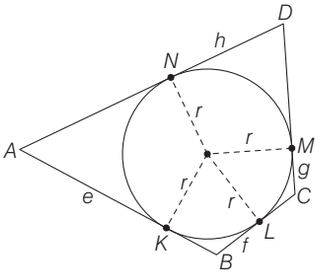
1. In a tangential quadrilateral, the four angle bisectors meet at the center of the incircle.
2. Conversely, a convex quadrilateral in which the four angle bisectors meet at a point must be tangential and the common point is the incenter.



3. The two pairs of opposite sides in a tangential quadrilateral add up to the same total length, which equals the semi perimeter s of the quadrilateral : $a + c = b + d = s$.
4. Conversely, a convex quadrilateral in which $a + c = b + d$ must be tangential.
5. A convex quadrilateral $ABCD$ is tangential, if and only if the incircles in the two triangles ABC and ADC are tangent to each other.



6. Area of the tangential quadrilateral —



(i) $A = r \times s$; where r is the inradius and s is the semi-perimeter

(ii) $A = \sqrt{(e+f+g+h)(efg+fgh+ghe+hef)}$; where, e, f, g, h are the lengths of the tangents.

(iii) $A = \frac{1}{2}\sqrt{p^2q^2 - (ac-bd)^2}$; where p, q are the diagonals and a, b, c, d are the sides of tangential quadrilateral.

(iv) $A = \sqrt{abcd - (eg-fh)^2}$; where a, b, c, d are the sides and e, f, g, h are the successive tangent lengths.

(v) When the tangential quadrilateral is cyclic, that is bi-centric, then $eg = fh$ and so the tangential quadrilateral will have the maximum area $\sqrt{abcd}$.

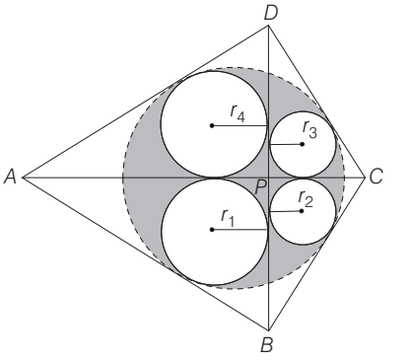
NOTE In the above diagram, it is considered as $AB = a, BC = b, CD = c, DA = d$ and $AK = e, BL = f, CM = g, DN = h$.

7. The maximum area of a tangential quadrilateral $A \le \sqrt{abcd}$
8. The minimum area of a tangential quadrilateral $A \ge 4r^2$
9. The minimum value of the semi-perimeter $s \ge 4r$
10. The inradius of a tangential quadrilateral $r = A/s$; where A is the area and s is the semi-perimeter of the tangential quadrilateral.
11. If e, f, g and h are the tangent lengths from A, B, C and D, respectively, to the points where the incircle is tangent to the sides of a tangential quadrilateral $ABCD$, then the lengths of the diagonals $p = AC$ and $q = BD$ are

$$p = \sqrt{\left(\frac{e+g}{f+h}\right)[(e+g)(f+h)+4fh]}$$

and $$q = \sqrt{\left(\frac{f+h}{e+g}\right)[(e+g)(f+h)+4eg]}$$

12. In the non-overlapping triangles APB, BPC, CPD, DPA formed by the diagonals in a convex quadrilateral $ABCD$, where the diagonals intersect at P, there are the following characterizations of tangential quadrilaterals.



(i) Let r_1, r_2, r_3 and r_4 denote the radii of the incircles in the four triangles APB, BPC, CPD and DPA respectively, then the quadrilateral is tangential if and only if $\frac{1}{r_1}+\frac{1}{r_3}=\frac{1}{r_2}+\frac{1}{r_4}$

(ii) If h_1, h_2, h_3 and h_4 denote the altitudes in the four triangles APB, BPC, CPD and DPA respectively, (from the diagonal intersection to the sides of the quadrilateral), then quadrilateral is tangential if and only if

$$\frac{1}{h_1}+\frac{1}{h_3}=\frac{1}{h_2}+\frac{1}{h_4}$$

(iii) If the exradii r_a, r_b, r_c and r_d in the four triangles APB, BPC, CPD and DPA respectively, (the four excircles are each tangent to one side of the quadrilateral and the extensions of its diagonals). A quadrilateral is tangential if and only if

$$\frac{1}{r_a}+\frac{1}{r_c}=\frac{1}{r_b}+\frac{1}{r_d}$$

(iv) If R_1, R_2, R_3 and R_4 denote the radii in the circumcircles of triangles APB, BPC, CPD and DPA, respectively, then the quadrilateral $ABCD$ is tangential if and only if

$$R_1+R_3=R_2+R_4$$

(v) A convex quadrilateral $ABCD$, with diagonals intersecting at P, is tangential if and only if the four excenters in triangles APB, BPC, CPD and DPA opposite the vertices B and D are concyclic. If R_a, R_b, R_c and R_d are the exradii in the triangles APB, BPC, CPD and DPA, respectively, opposite the vertices B and D, then another condition is that the quadrilateral is tangential if and only if

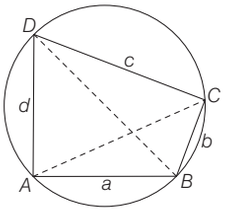
$$\frac{1}{R_a}+\frac{1}{R_c}=\frac{1}{R_b}=\frac{1}{R_d}$$

(vi) A convex quadrilateral $ABCD$ with diagonals intersecting at P is tangential if and only if

$$\frac{a}{\Delta(APB)}+\frac{c}{\Delta(CPD)}=\frac{b}{\Delta(BPC)}+\frac{d}{\Delta(DPA)}$$

Cyclic Quadrilateral

A cyclic quadrilateral is a quadrilateral for which a circle can be circumscribed so that it touches each vertex of the quadrilateral.



(i) Semi perimeter $s=\frac{a+b+c+d}{2}$

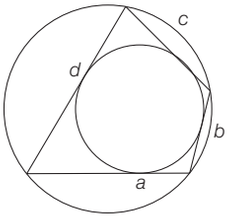
(ii) Area $A=\sqrt{(s-a)(s-b)(s-c)(s-d)}$

(iii) Area $A=\frac{\sqrt{(ab+cd)(ac+bd)(ad+bc)}}{4R}$

(iv) Circumradius $R=\frac{1}{4}\sqrt{\frac{(ab+cd)(ac+bd)(ad+bc)}{(s-a)(s-b)(s-c)(s-d)}}$

Bicentric Quadrilateral

A quadrilateral is said to be bicentric if it is cyclic and tangential.



(i) If the sides be a, b, c, d the area of a bicentric quadrilateral is $A=\sqrt{abcd}$

(ii) Inradius, $r=\frac{2\sqrt{abcd}}{a+b+c+d}=\frac{\sqrt{abcd}}{s}$

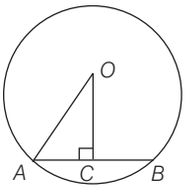
(iii) Circumradius, $R=\frac{1}{4}\sqrt{\frac{(ab+cd)(ac+bd)(ad+bc)}{abcd}}$

(iv) In a bicyclic quadrilateral, the distance x between the incentre and circumcentre is given by

$$\frac{1}{(R-x)^2}+\frac{1}{(R+x)^2}=\frac{1}{(r)^2}$$

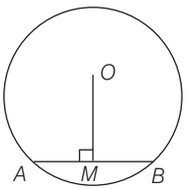
Introductory *Exercise 12.5*

1. In the given figure, O is the centre of the circle. Radius of the circle is 17 cm.If $OC = 8$ cm, then the length of the chord AB is :



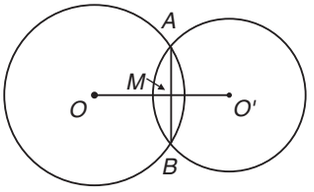
(a) 35 cm (b) 30 cm (c) 15 cm (d) 18 cm

2. In the given figure $OM \perp AB$, radius of the circle is 5 cm and length of the chord $AB = 8$ cm. Find the measure of OM.



(a) 3 cm (b) 2.5 cm (c) 2 cm (d) 6 cm

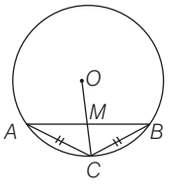
3. In the given figure, two circles with their respective centres intersect each other at A and B and AB intersects OO' at M, then $m\angle OMA$ is :



(a) 60° (b) 80°
(c) 90° (d) can't be determined

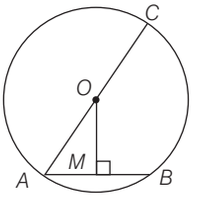
4. In the above question (no. 3) what is the ratio of $AM : BM$?
(a) 5 : 6 (b) 3 : 2
(c) 1 : 1 (d) can't be determined

5. In the given figure the two chords AC and BC are equal. The radius OC intersect AB at M, then $AM : BM$ is :



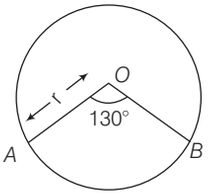
(a) 1 : 1 (b) $\sqrt{2} : 3$
(c) $3 : \sqrt{2}$ (d) none of these

6. In the adjoining figure, O is the centre of circle and diameter $AC = 26$ cm. If chord $AB = 10$ cm, then the distance between chord AB and centre O of the circle is :



(a) 24 cm (b) 16 cm
(c) 12 cm (d) none of these

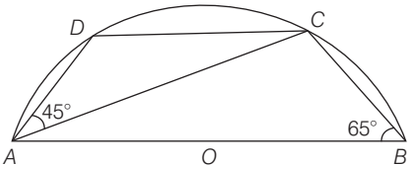
7. In the adjoining circle $C(0, r)$ the degree measure of minor arc $AB = 130°$. Find the degree measure of major arc.



(a) 230° (b) 260°
(c) 310° (d) none of these

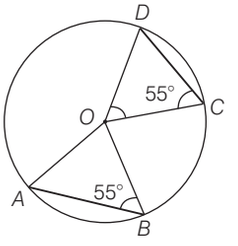
8. If the diagonals of a cyclic quadrilateral are equal, then the quadrilateral is :
(a) rhombus (b) square
(c) rectangle (d) none of these

9. In the given figure, AB is diameter of the circle. C and D lie on the semicircle. $\angle ABC = 65°$ and $\angle CAD = 45°$. Find $m\angle DCA$.



(a) 45° (b) 25°
(c) 20° (d) none of these

10. In the given figure, chords AB and CD are equal. If $\angle OBA = 55°$, then $m\angle COD$ is :

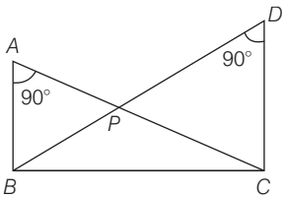


(a) 65° (b) 55° (c) 70° (d) 50°

11. The quadrilateral formed by angle bisectors of a cyclic quadrilateral is a :

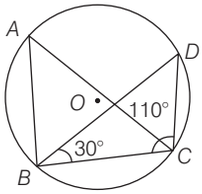
(a) rectangle (b) square
(c) parallelogram (d) cyclic quadrilateral

12. ΔABC and ΔDBC have a common base and drawn towards one sides. $\angle BAC = \angle BDC = 90°$. If AC and DB intersect at P, then :



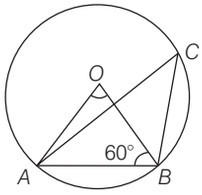
(a) $AP \times PC = BP \times PD$
(b) $AP \times BP = PC \times PD$
(c) $AP \times PD = PC \times BP$
(d) none of the above

13. In the given figure, $\angle BAC$ and $\angle BDC$ are the angles of same segments. $\angle DBC = 30°$ and $\angle BCD = 110°$. Find $m\ \angle BAC$.



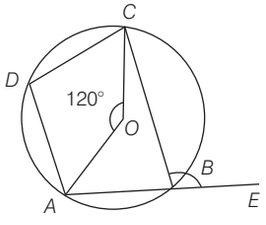
(a) 35° (b) 40°
(c) 55° (d) 60°

14. In the given figure, O is the centre of the circle, $\angle ABO = 60°$. Find the value of $\angle ACB$.



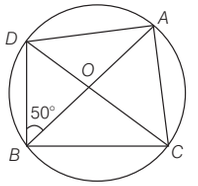
(a) 40° (b) 60°
(c) 50° (d) 30°

15. In the given figure, $\angle AOC = 120°$. Find $m\ \angle CBE$, where O is the centre.



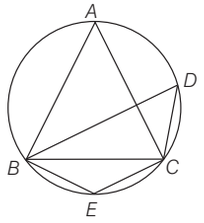
(a) 60° (b) 100° (c) 120° (d) 150°

16. In the adjoining figure, O is the centre of the circle and $\angle OBD = 50°$. Find the $m\ \angle BAD$.



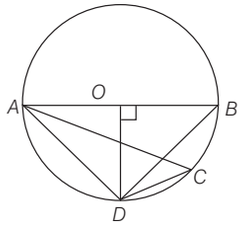
(a) 60° (b) 40° (c) 80° (d) 45°

17. In the given figure, ΔABC is an equilateral triangle. Find $m\ \angle BEC$.



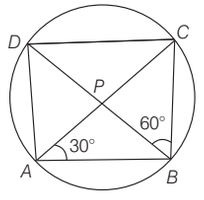
(a) 120° (b) 60°
(c) 80° (d) none of these

18. In the given figure, AB is the diameter of the circle. Find the value of $\angle ACD$.



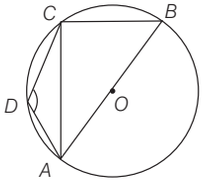
(a) 30° (b) 60° (c) 45° (d) 25°

19. In the given figure, $ABCD$ is a cyclic quadrilateral and diagonals bisect each other at P. If $\angle DBC = 60°$ and $\angle BAC = 30°$, then $\angle BCD$ is :



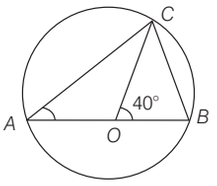
(a) 90° (b) 60°
(c) 80° (d) none of these

20. In the given figure, $ABCD$ is a cyclic quadrilateral and AB is the diameter. $\angle ADC = 140°$, then find $m\ \angle BAC$.



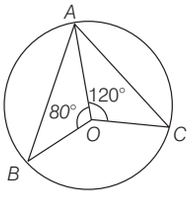
(a) 45° (b) 40° (c) 50° (d) none of these

21. In the given figure, $\angle COB = 40°$, AB is the diameter of the circle. Find $m\ \angle CAB$.



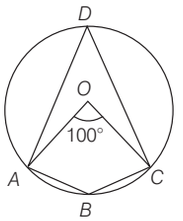
(a) 40° (b) 20° (c) 30° (d) none of these

22. In the given figure, O is the centre of circle. $\angle AOB = 80°$ and $\angle AOC = 120°$. Find $m\ \angle BAC$.



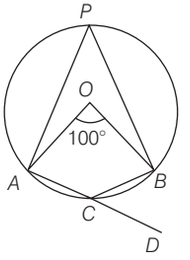
(a) 120° (b) 80° (c) 100° (d) none of these

23. In the given figure, O is the centre of the circle and $\angle AOC = 100°$. Find the ratio of $m\ \angle ADC : m\ \angle ABC$.



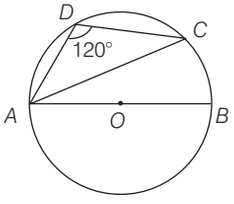
(a) 5 : 6 (b) 1 : 2
(c) 5 : 13 (d) none of these

24. In the given figure, O is the centre of circle, $\angle AOB = 100°$. Find $m\ \angle BCD$.



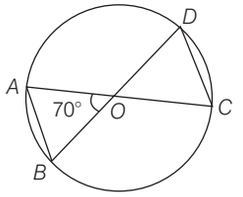
(a) 80° (b) 60° (c) 50° (d) 40°

25. In the given figure, AB is the diameter of the circle. $\angle ADC = 120°$. Find $m\ \angle CAB$.



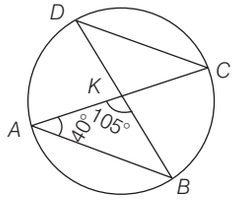
(a) 20° (b) 30°
(c) 40° (d) can't be determined

26. In the given figure, O is the centre of the circle. $\angle AOB = 70°$, find $m\ \angle OCD$.



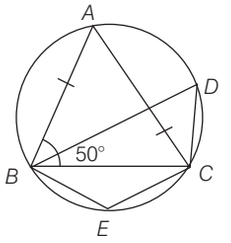
(a) 70° (b) 55°
(c) 65° (d) 110°

27. In the given figure, $\angle CAB = 40°$ and $\angle AKB = 105°$. Find $\angle KCD$.



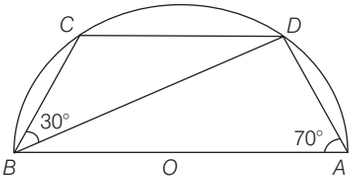
(a) 65° (b) 35° (c) 40° (d) 72°

28. In the given figure, ABC is an isosceles triangle in which $AB = AC$ and $m\ \angle ABC = 50°$. Find $m\ \angle BDC$.



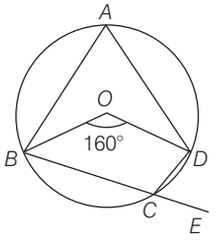
(a) 80° (b) 60° (c) 65° (d) 100°

29. In the given figure, AB is the diameter. $m\ \angle BAD = 70°$ and $m\ \angle DBC = 30°$. Find $m\ \angle BDC$.



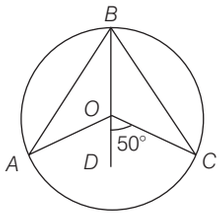
(a) 25° (b) 30° (c) 40° (d) 60°

30. Find the value of $\angle DCE$.



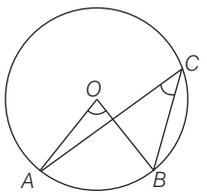
(a) 100° (b) 80°
(c) 90° (d) 75°

31. '*O*' is the centre of the circle, line segment *BOD* is the angle bisector of $\angle AOC$, $m\angle COD = 50°$.
Find $m\angle ABC$.



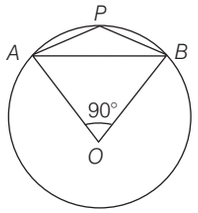
(a) 25° (b) 50°
(c) 100° (d) 120°

32. In the given figure, *O* is the centre of the circle and $\angle ACB = 25°$. Find $\angle AOB$.



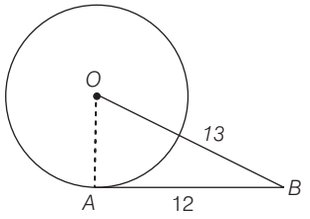
(a) 25° (b) 50°
(c) 75° (d) 60°

33. In the given figure, *O* is the centre of the circle. $\angle AOB = 90°$. Find $m\angle APB$.



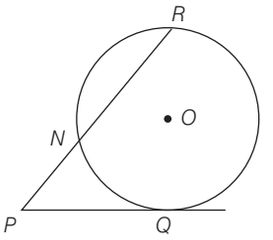
(a) 130°
(b) 150°
(c) 135°
(d) can't be determined

34. In the given figure, *O* is the centre of the circle. *AB* is tangent. *AB* = 12 cm and *OB* = 13 cm. Find *OA*.



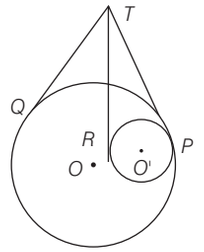
(a) 6.5 cm
(b) 6 cm
(c) 5 cm
(d) none of the above

35. In the given figure, *PQ* is the tangent of the circle. Line segment *PR* intersects the circle at *N* and *R*. *PQ* = 15 cm, *PR* = 25 cm, find *PN*.



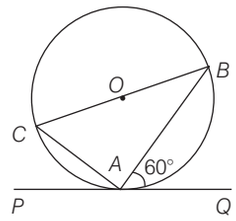
(a) 15 cm (b) 10 cm (c) 9 cm (d) 6 cm

36. In the given figure, there are two circles with the centres *O* and *O'* touching each other internally at *P*. Tangents *TQ* and *TP* are drawn to the larger circle and tangents *TP* and *TR* are drawn to the smaller circle. Find *TQ* : *TR*.



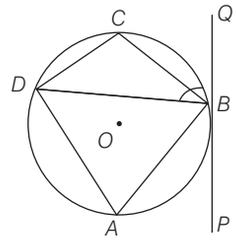
(a) 8 : 7 (b) 7 : 8
(c) 5 : 4 (d) 1 : 1

37. In the given figure, *PAQ* is the tangent. *BC* is the diameter of the circle. $m\angle BAQ = 60°$, find $m\angle ABC$.



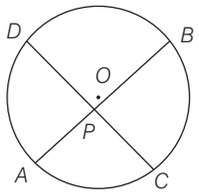
(a) 25° (b) 30° (c) 45° (d) 60°

38. *ABCD* is a cyclic quadrilateral *PQ* is a tangent at *B*. If $\angle DBQ = 65°$, then $\angle BCD$ is :



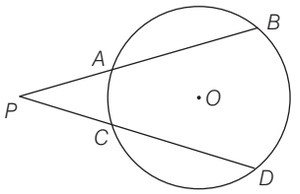
(a) 35° (b) 85° (c) 115° (d) 90°

39. In the given figure, $AP = 2$ cm, $BP = 6$ cm and $CP = 3$ cm. Find DP.



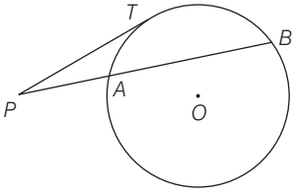
(a) 6 cm (b) 4 cm
(c) 2 cm (d) 3 cm

40. In the given figure, $AP = 3$ cm, $BA = 5$ cm and $CP = 2$ cm. Find CD.



(a) 12 cm (b) 10 cm
(c) 9 cm (d) 6 cm

41. In the given figure, tangent $PT = 5$ cm, $PA = 4$ cm, find AB.

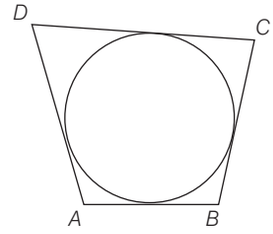


(a) $\frac{7}{4}$ cm
(b) $\frac{11}{4}$ cm
(c) $\frac{9}{4}$ cm
(d) can't be determined

42. Two circles of radii 13 cm and 5 cm touch internally each other. Find the distance between their centres.
(a) 18 cm (b) 12 cm
(c) 9 cm (d) 8 cm

43. Three circles touch each other externally. The distance between their centre is 5 cm, 6 cm and 7 cm. Find the radii of the circles.
(a) 3 cm, 2 cm, 4 cm
(b) 3 cm, 4 cm, 1 cm
(c) 1cm, 2.5 cm, 3.5 cm
(d) 1 cm, 2 cm, 4 cm

44. A circle touches a quadrilateral $ABCD$. Find the true statement

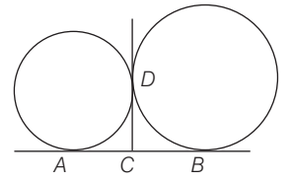


(a) $AB + BC = CD + AD$ (b) $AB + CD = BC + AD$
(c) $BD = AC$ (d) none of these

45. O and O' are the centres of two circles which touch each other externally at P. AB is a common tangent. Find $\angle APO$.
(a) 90° (b) 120°
(c) 60° (d) data insufficient

46. If AB is a chord of a circle, P and Q are two points on the circle different from A and B, then :
(a) the angle subtended by AB at P and Q are either equal or supplementary.
(b) the sum of the angles subtended by AB at P and Q is always equal two right angles.
(c) the angles subtended at P and Q by AB are always equal.
(d) the sum of the angles subtended at P and Q is equal to four right angles.

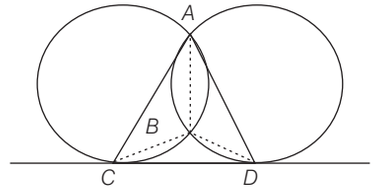
47. In the given figure, AB and CD are two common tangents to the two touching circle. If $CD = 6$ cm, then AB is equal to :



(a) 9 cm (b) 15 cm
(c) 12 cm (d) none of these

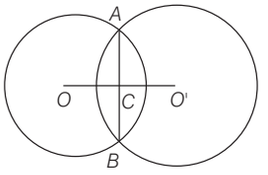
48. In the given figure, CD is a direct common tangent to two circles intersecting each other at A and B, then :

$\angle CAD + \angle CBD = ?$



(a) 120° (b) 90°
(c) 360° (d) 180°

49. O and O' are the centres of circle of radii 20 cm and 37 cm. AB = 24 cm. What is the distance OO' ?



(a) 51 cm (b) 45 cm
(c) 35 cm (d) 48 cm

50. In a circle of radius 5 cm, AB and AC are the two chords such that $AB = AC = 6$ cm. Find the length of the chord BC.
(a) 4.8 cm (b) 10.8 cm
(c) 9.6 cm (d) none of these

51. In a circle of radius 17 cm, two parallel chords are drawn on opposite sides of a diameter. The distance between the chords is 23 cm. If the length of one chord is 16 cm, then the length of the other is :
(a) 23 cm (b) 30 cm
(c) 15 cm (d) none of these

52. If two circles are such that the centre of one lies on the circumference of the other, then the ratio of the common chord of two circles to the radius of any of the circles is :
(a) $\sqrt{3} : 2$
(b) $\sqrt{3} : 1$
(c) $\sqrt{5} : 1$
(d) none of the above

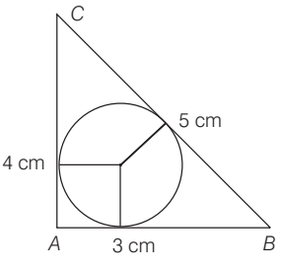
53. Two circles touch each other internally. Their radii are 2 cm and 3 cm. The biggest chord of the other circle which is outside the inner circle, is of length :
(a) $2\sqrt{2}$ cm (b) $3\sqrt{2}$ cm
(c) $2\sqrt{3}$ cm (d) $4\sqrt{2}$ cm

54. Through any given set of four points P, Q, R, S it is possible to draw :
(a) atmost one circle
(b) exactly one circle
(c) exactly two circles
(d) exactly three circles

55. The distance between the centres of equal circles each of radius 3 cm is 10 cm. The length of a transverse tangent is :
(a) 4 cm (b) 6 cm (c) 8 cm (d) 10 cm

56. The number of common tangents that can be drawn to two given circles is at the most :
(a) 1 (b) 2 (c) 3 (d) 4

57. ABC is a right angled triangle AB = 3 cm, BC = 5 cm and AC = 4 cm, then the inradius of the circle is :



(a) 1 cm (b) 1.25 cm
(c) 1.5 cm (d) none of these

58. A circle has two parallel chords of lengths 6 cm and 8 cm. If the chords are 1 cm apart and the centre is on the same side of the chords, then a diameter of the circle is of length :
(a) 5 cm (b) 6 cm
(c) 8 cm (d) 10 cm

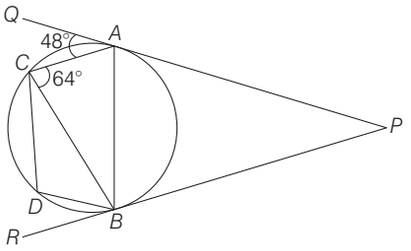
59. Three equal circles of unit radius touch each other. Then, the area of the circle circumscribing the three circles is :
(a) $6\pi\ (2 + \sqrt{3})^2$ (b) $\frac{\pi}{6}(2 + \sqrt{3})^2$
(c) $\frac{\pi}{3}(2 + \sqrt{3})^2$ (d) $3\pi\ (2 + \sqrt{3})^2$

60. The radius of a circle is 20 cm. The radii (in cm) of three concentric circles drawn in such a manner that the whole area is divided into four equal parts, are :
(a) $20\sqrt{2}, 20\sqrt{3}, 20$
(b) $\frac{10\sqrt{3}}{3}, \frac{10\sqrt{2}}{3}, \frac{10}{3}$
(c) $10\sqrt{3}, 10\sqrt{2}, 10$
(d) 17, 14, 9

61. Points A, B, C and D are concyclic. BC is the diameter and O is the centre of the circle, $\angle BAO = 69°$, find $\angle DAO + \angle DCO$.
(a) 138 (b) 111
(c) 121 (d) none of these

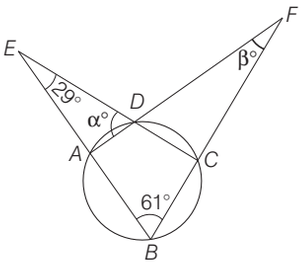
62. An isosceles right triangle ABC has an incircle with centre O, which touches the sides AB, BC and AC at D, E and F respectively. If the longest side AC is $\sqrt{2}$ unit, find DE.
(a) $\sqrt{2} - 1$ cm (b) $\sqrt{2} - \frac{1}{\sqrt{2}}$ cm
(c) $1 - \frac{1}{\sqrt{2}}$ cm (d) none of these

63. Consider the following diagram. In this diagram, *A*, *B*, *D* and *C* are cyclic. *QP* and *RP* are the tangents at *A* and *B* respectively and they meet at *P*. $\angle CAQ = 48°$ and $ACB = 64°$. Find the value of $\angle BDC + \angle CBP + \angle APB$.



(a) 248° (b) 276°
(c) 360° (d) none of these

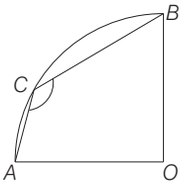
64. In the following figure *A*, *B*, *C* and *D* are cyclic. *BE*, *CE*, *AF* and *BF* are straight lines



Find the value of $\alpha - \beta$.
(a) 28° (b) 41°
(c) 32° (d) none of these

65. In a circle with centre *O*, there are two chords *AB* and *BC*, such that $\angle OAB = 24°$ and $\angle OCB = 42°$. Find the measure of $\angle AOC$.
(a) 36° (b) 63°
(c) 228° (d) either (a) or (c)

66. In the following figure, *AOBC* is a quadrilateral inscribed in a quadrant of a circle such that $\angle AOB$ is a right angle and point *C* falls on the arc of the quadrant. Find $\angle ACB$.



(a) 120° (b) 135°
(c) 150° (d) data insufficient

67. The sides of a triangle ΔABC are 15 cm, 41 cm and 52 cm. Find the diameter of the circumcircle of the ΔABC.
(a) $\frac{205}{3}$ cm (b) $\frac{208}{3}$ cm
(c) 69 cm (d) 70 cm

68. The radius of a circle is 2 cm, what is its curvature?
(a) 1 cm (b) 4 cm
(c) 0.5 cm (d) 1.414 cm

69. The curvatures of two externally tangent circles are 0.5 and 2, what is the distance between their centers?
(a) 2.5 (b) 1
(c) 4 (d) 8

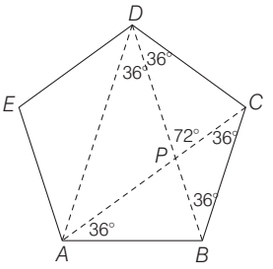
70. An equilateral triangle *ABC* is inscribed in a circle and a chord *AD* is connected to two other chords *BD* and *CD*, whereas point *D* is on the arc *BC*. If the side of the triangle is 30 cm and the chord *AD* is 32 cm, find *BD* + *CD*.
(a) 31 cm (b) 32 cm
(c) 225/8 cm (d) 256/15 cm

13 In ΔACB, $\angle ABC = 108°$ and $AB = BC$,

$\therefore \quad \angle BAC = \angle BCA = 36°$

Similarly, in ΔDCB, $\angle DCB = 108°$ and $CD = CB$,

$\therefore \ \angle CDB = \angle CBD = 36°$



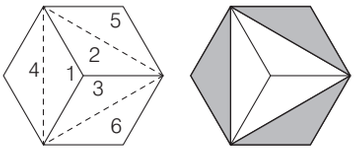
Now, in ΔBPC, $\angle PBC = \angle PCB = 36°$,

$\therefore \quad \angle BPC = 180 - (36 + 36) = 108°$

Thus, $\angle DPC = 180 - \angle BPC = 180 - 108 = 72°$

Hence choice (b) is the correct one.

14 Connect the centre of the hexagon with the three vertices of the given triangle, as shown below. Due to symmetricity there are now total 6 congruent triangles, with 30-30-120 degree angles. Out of 6 such triangles exactly 3 triangles are comprised in the given triangle.



So the required ratio $= \frac{3}{6} = \frac{1}{2}$

Hence choice (d) is the correct one.

☑ Introductory Exercise 12.5

1 $AC = \sqrt{OA^2 - OC^2} = \sqrt{(17)^2 - (8^2)} = 15\,\text{cm}$

$\therefore \quad AB = 2AC = 30\,\text{cm} \qquad (\because AC = BC)$

2 $AB = 8\,\text{cm}$

$\therefore \quad AM = 4\,\text{cm}$

and $\quad OA = 5\,\text{cm}$

$\therefore \quad OM = \sqrt{(OA)^2 - (AM)^2} = \sqrt{25 - 16} = 3\,\text{cm}$

3 $\because AB$ is the perpendicular bisector of OO'.

$\therefore \quad \angle OMA = 90°$

4 $AM : BM : 1 : 1$

5 $\angle AOC = \angle BOC \qquad (\because AC = BC)$

$\therefore OC$ is the perpendicular bisector of AB.

$\therefore \quad AM = BM$

6 $OA = 13\,\text{cm} \Rightarrow AM = 5\,\text{cm}$

$\therefore \quad OM = \sqrt{(13)^2 - (5)^2} = 12\,\text{cm}$

7 Degree measure of major arc $= 360° - 130° = 230°$

8 It is necessarily a rectangle.

9 $\angle ADC = 180° - 65° = 115°$

$\therefore \quad \angle DCA = 180° - (115° + 45°) = 20°$

($ADCB$ is a cyclic quadrilateral)

10 $\angle OBA = \angle OAB = 55°$

$\therefore \quad \angle AOB = 180° - (55° + 55°) = 70°$

and $\quad \angle AOB = \angle COD = 70°$

12 $\Delta BAP \sim \Delta CDP \quad (\because \angle A = \angle D$ and $\angle APB = \angle DPC)$

$\therefore \quad \frac{AB}{CD} = \frac{AP}{DP} = \frac{BP}{CP}$

$\therefore \quad AP \times PC = BP \times DP$

13 $\angle BDC = 180° - (110° + 30°) = 40°$

and $\quad \angle BAC = \angle BDC = 40°$ (angles of the same segment)

14 $\angle ABO = 60° = \angle BAO \ (\because AO = BO$, radii of the circle)

$\therefore \quad \angle AOB = 180° - (60° + 60°) = 60°$

$\therefore \quad \angle ACB = \frac{1}{2} \angle AOB = 30°$

15 $\angle CBA = \frac{1}{2} \angle COA = 60°$

$\therefore \quad \angle CBE = 180° - \angle CBA$

$180° - 60° = 120°$

16 $\angle BDA = 90° \Rightarrow \angle DBA = 50°$

$\therefore \quad \angle DAB = 180° - (90° + 50°) = 40°$

17 $\angle BAC = 60°$

$\therefore \quad \angle BEC = 180° - 60° = 120°$

18 $\angle ABD = 45° \quad \therefore \ \angle ACD = 45°$

19 $\angle BDC = \angle BAC = 30°$

$\therefore \ \angle BCD + \angle BDC + \angle DBC = 180°$

$\therefore \quad \angle BCD = 180° - (30° + 60°)$

$\angle BCD = 90°$

20 $\angle ADC = 140°$

$\therefore \quad \angle ABC = 40° \Rightarrow \angle ACB = 90°$

$\therefore \quad \angle BAC = 180° - (90° + 40°) = 50°$

21 $\angle CAB = \frac{1}{2} \angle BOC = \frac{1}{2} \times 40° = 20°$

22 $\angle OBA = \angle OAB = \left(\frac{180° - 80°}{2}\right) = 50°$

and $\quad \angle OCA = \angle OAC = \left(\frac{180° - 120°}{2}\right) = 30°$

$\therefore \quad \angle BAC = \angle BAO + \angle OAC = 50° + 30° = 80°$

23 $\angle ADC = \frac{100°}{2} = 50°$

$\therefore \quad \angle ABC = 180° - 50° = 130°$

$\therefore \quad \frac{m\,\angle ADC}{m\,\angle ABC} = \frac{50°}{130°} = \frac{5}{13}$

24 $\angle APB = 50^\circ \quad \left(\because m\angle APB = \frac{1}{2} m\angle AOB\right)$

$\therefore \quad \angle ACB = 180^\circ - 50^\circ = 130^\circ$

$\therefore \quad \angle BCD = 180^\circ - 130^\circ = 50^\circ$

25 $\angle ABC = 180^\circ - 120^\circ = 60^\circ$

$\angle ACB = 90^\circ$

$\therefore \quad \angle CAB = 180^\circ - (90^\circ + 60^\circ) = 30^\circ$

26 $\angle COD = \angle AOB = 70^\circ$

$\therefore \quad \angle OCD = \frac{180^\circ - 70^\circ}{2} = 55^\circ$

27 $\angle ABK = 180^\circ - (105^\circ + 40^\circ) = 35^\circ$

$\therefore \quad \angle ACD = \angle ABD = 35^\circ$

28 $\angle BAC = 80^\circ \quad (\because \angle ABC = \angle ACB = 50^\circ)$

$\therefore \quad \angle BDC = \angle BAC = 80^\circ$

29 $\angle ADB = 90^\circ$

$\therefore \quad \angle DBA = 20^\circ \quad [180^\circ - (90^\circ + 70^\circ)]$

$\therefore \quad \angle CBA = 30^\circ + 20^\circ = 50^\circ$

$\therefore \quad \angle CDA = 180^\circ - 50^\circ = 130^\circ$

$\therefore \quad \angle CDB = 130^\circ - 90^\circ = 40^\circ$

30 $\angle BAD = 80^\circ \ \therefore \ \angle DCE = 80^\circ$

31 $\angle AOC = 2 \times 50 = 100^\circ$

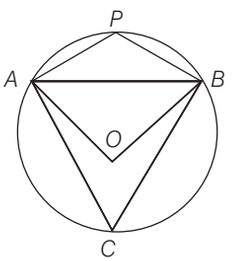
$\therefore \quad \angle ABC = \frac{100}{2} = 50^\circ$

NOTE There are some more ways to solve it. Find them.

32 $\angle AOB = 2\angle ACB = 2 \times 25 = 50^\circ$

33 $\angle APB = 180^\circ - \frac{1}{2} \times 90^\circ = 135^\circ$

$\left[\because \angle ACB = 45^\circ = \left(\frac{1}{2} \times 90^\circ\right) \right.$
$\left. \text{and } \angle APB + \angle ACB = 180^\circ \right]$



34 $OA = \sqrt{(OB)^2 - (AB)^2}$

$\therefore \quad OA = \sqrt{169 - 144} = 5\text{ cm}$

35 $PN \times PR = PQ^2 \Rightarrow PN \times 25 = (15)^2$

$\therefore \quad PN = 9\text{ cm}$

36 $TQ = TP \quad \text{and} \quad TP = TR$

$\therefore \quad TQ = TP = TR$

37 $\angle BAC = 90^\circ$

$\angle BCA = 60^\circ \quad (\because \angle BCA = BAQ)$

$\therefore \quad \angle ABC = 180^\circ - (90^\circ + 60^\circ)$

$\angle ABC = 30^\circ$

38 $\angle DBQ = 65^\circ$

$\therefore \quad \angle DAB = 65^\circ$

$\therefore \quad \angle DCB = 180^\circ - 65^\circ = 115^\circ$

39 $AP \times BP = DP \times CP$

$2 \times 6 = DP \times 3$

$\Rightarrow \quad DP = 4\text{ cm}$

40 $PA \times PB = PC \times PD$

$3 \times (3 + 5) = 2 \times PD$

$\therefore \quad PD = 12\text{ cm}$

$\therefore \quad CD = PD - PC = 12 - 2 = 10\text{ cm}$

41 $PT^2 = PA \times PB$

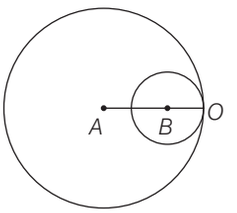
$5 \times 5 = 4 \times (4 + x)$

$\Rightarrow \quad x = \frac{9}{4}\text{ cm}, \qquad \text{where } AB = x$

42 Let they touch eachother at O and A, B be the centres of the two circles, then $OA = 13\text{ cm}$

and $OB = 5\text{ cm}$

$\therefore \quad AB = OA - OB = 8\text{ cm}$



43 $AB = 5\text{ cm} = x + y$

$BC = 6\text{ cm} = y + z$

$AC = 7\text{ cm} = z + x$

$\therefore \quad 2(x + y + z) = 5 + 6 + 7 = 18\text{ cm}$

$\Rightarrow \quad x + y + z = 9\text{ cm}$

$\Rightarrow x = (x + y + z) - (y + z) = 9 - 6 = 3\text{ cm}$

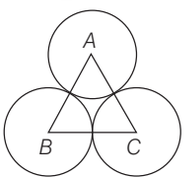
$\therefore \quad x + y = 5\text{ cm}$

$\therefore \quad y = 5 - x = 5 - 3 = 2\text{ cm}$

$\therefore \quad z = 6 - y = 6 - 2 = 4\text{ cm}$

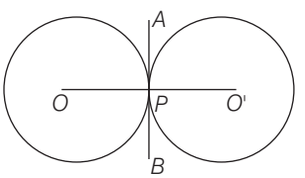
$\therefore \quad x = 3\text{ cm}$

$y = 2\text{ cm} \text{ and } z = 4\text{ cm}$



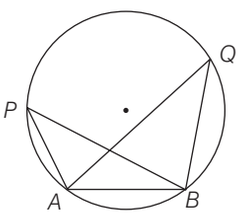
44 Prove with the help of property of tangents : "Tangents drawn from the same point on a circle are always equal."

45 Tangent is always perpendicular to the radius.



46 $\angle APB = \angle AQB$

when $\angle APB = \angle AQB = 90^\circ$



then they are supplementary. Also they are supplementary when they are in different segments.

47 $CD = 6$ cm

$\therefore$ $AC = 6$ cm and $BC = 6$ cm

($\because AC = CD$ and $BC = CD$. Two tangents from the same point are always equal.)

$\therefore$ $AB = 12$ cm

48 $\angle CAB = \angle BCD$

and $\angle DAB = \angle BDC$

$\therefore$ $\angle CAD = \angle CAB + \angle DAB = \angle BCD + \angle BDC$

$\Rightarrow$ $\angle CAD = \angle BCD + \angle BDC$

$\therefore$ $\angle CAD + \angle CBD = \angle BCD + \angle BDC + \angle CBD = 180°$

49 $AB = 24$ cm

$\therefore$ $AC = BC = 12$ cm

$OA = 20$ cm

$\therefore$ $OC = \sqrt{(OA)^2 - (AC)^2}$

$= \sqrt{400 - 144} = 16$ cm

and $O'A = 37$ cm

$\therefore$ $O'C = \sqrt{(O'A)^2 - (AC)^2}$

$= \sqrt{(37)^2 - (12)^2} = 35$ cm

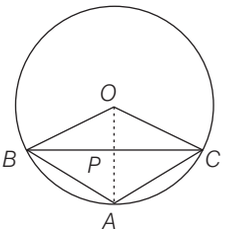
$\therefore$ $OO' = OC + CO' = 16 + 35 = 51$ cm

50 $AB = AC$, $OB = OC$

and BC is common and OA is common

also $\angle BOA = \angle COA$

$\therefore$ $\Delta BOA \cong \Delta COA$



$\therefore$ BC and OA are perpendicular bisector to each other

[Area of $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$]

$$s = \frac{a+b+c}{2} = \frac{5+5+6}{2} = 8 \text{ cm}$$

Area of $\Delta BOA = \sqrt{8 \times 3 \times 3 \times 2} = 12 \text{ cm}^2$

Again area of $\Delta BOA = \frac{1}{2} OA \times BP = \frac{1}{2} \times 5 \times BP$

$$12 = \frac{1}{2} \times 5 \times BP$$

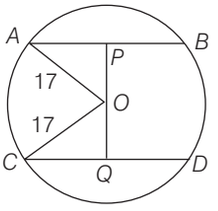
$\Rightarrow$ $BP = 4.8$ cm

$\therefore$ $BC = 2BP = 2 \times 4.8 = 9.6$ cm

51 $OA = OC = 17$ cm

$$AP = \frac{1}{2} AB = \frac{1}{2} \times 16 = 8 \text{ cm}$$

$\therefore$ $OP = \sqrt{OA^2 - AP^2} = 15$ cm



Now, $PQ = OP + OQ = 23$ cm

$\therefore$ $15 + OQ = 23$ cm

$\therefore$ $OQ = 8$ cm

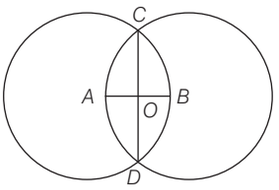
Again, $CQ = \sqrt{(OC)^2 - (OQ)^2} \Rightarrow CQ = \sqrt{(17)^2 - (8)^2}$

$\therefore$ $CQ = 15$ cm

$\therefore$ $CD = 2CQ = 30$ cm

52 $AB = r$ (say) then $AC = BC = r$, also

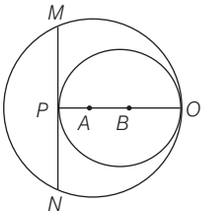
$\therefore$ $OA = OB = r/2$ (CD is a common chord)



$$\therefore OC = \sqrt{(AC)^2 - (OA)^2} = \sqrt{r^2 - \left(\frac{r}{2}\right)^2} = \sqrt{\frac{3r^2}{4}} = \frac{\sqrt{3}}{2} r$$

$$\therefore CD = 2CO = 2 \times \frac{\sqrt{3}}{2} r = \sqrt{3} r \therefore \frac{l(CD)}{l(AC)} = \frac{\sqrt{3} r}{r} = \frac{\sqrt{3}}{1}$$

53 Let A be the centre of larger circle and B be the centre of the smaller circle.



$OA = 3$ cm and $BO = 2$ cm

and $AB = (OA - OB) = 1$ cm

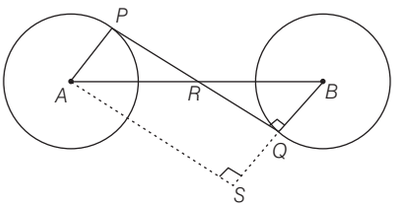
also $BP = 2$ cm $\therefore$ $AP = 1$ cm

P is the point of contact of tangent MN.

$\therefore$ $MP = \sqrt{AM^2 - AP^2} = \sqrt{3^2 - 1^2} = 2\sqrt{2}$ cm

$\therefore$ $MN = 2MP = 4\sqrt{2}$ cm

55. $PQ = AS$ (PQ is a transverse common tangent)



and $BS = BQ + SQ = BQ + AP$

$BS = 2BQ$ ($BQ = AP$ = radius)

$BS = 6$ cm

$\therefore$ $AS = \sqrt{(AB)^2 - (BS)^2} = \sqrt{(10)^2 - (6)^2} = 8$ cm

$\therefore$ $AS = 8$ cm $\therefore$ $PQ = 8$ cm

Alternatively The length of transverse common tangent

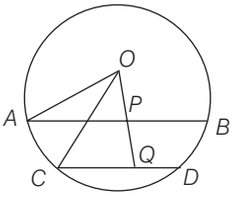
$= \sqrt{(\text{distance beween the centres})^2 - (r_1 + r_2)^2}$

$= \sqrt{(10)^2 - (3+3)^2} = 8$ cm

57 Inradius of an incircle of right angled triangle

$$= \frac{\text{Base} + \text{Altitude} - \text{Hypotenuse}}{2}$$

$$= \frac{4 + 3 - 5}{2} = 1 \text{ cm}$$



Alternatively

$$\text{Inradius} = \frac{\text{Area of triangle}}{\text{Semiperimeter of triangle}}$$

$$= \frac{(3 \times 4)/2}{(3 + 4 + 5)/2} = \frac{6}{6} = 1 \text{ cm}$$

58 Let AB and CD be the two chords, then $AB = 8$ cm and $CD = 6$ cm.

$\therefore$ $AP = 4$ cm and $CQ = 3$ cm and $PQ = 1$ cm (given)

Let $OP = x$

$\therefore$ $AO^2 = AP^2 + OP^2$

$\therefore$ $AO^2 = 4^2 + x^2 = 16 + x^2$ …(i)

and $OC^2 = CQ^2 + OQ^2$

$= CQ^2 + (OP + PQ)^2 = (3)^2 + (x + 1)^2$

$OC^2 = 9 + x^2 + 1 + 2x$ …(ii)

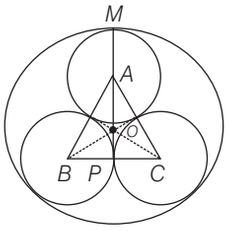
But $OA = OC$, are the radii of the same circle.

$\therefore$ $16 + x^2 = 9 + x^2 + 1 + 2x \Rightarrow x = 3$ cm

$\therefore$ $OA^2 = 16 + x^2 = 16 + 9 = 25$

$\therefore$ $OA = 5$ cm

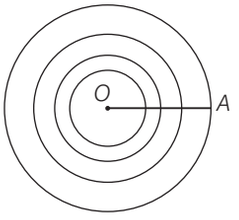
$\therefore$ Diameter of the circle $= 2(OA) = 10$ cm



59 $AB = BC = AC = 2$ cm

($\because$ radius of each circle = 1 cm)

$\therefore AP = \frac{\sqrt{3}}{2} \times 2 \Rightarrow AP = \sqrt{3}$ cm



Let O be the centroid, then

$$OA = \frac{2}{3} \times \sqrt{3} = \frac{2}{\sqrt{3}} \text{ cm}$$

and $AM = 1$ cm

$\therefore$ $OM = (2/\sqrt{3}) + 1 = 2 + \sqrt{3}/\sqrt{3}$ cm

OM is the radius of the larger circle.

$\therefore$ Area of the circumscribing circle

$$= \pi R^2 = \pi \times \left(\frac{2+\sqrt{3}}{\sqrt{3}}\right)^2 = \frac{\pi}{3}(2+\sqrt{3})^2$$

60 OA is the radius of the largest circle and O is the centroid of the all concentric circles. Let r_1, r_2, r_3 and r_4 be the radii of the concentric circles in increasing order then

Total area of given circle $= \pi \times (20)^2 = 400\pi$

Area of each region $= \frac{1}{4} \times 400\,\pi = 100\pi$

$\therefore$ Area of central region $= 100\pi = \pi r_1^2$

$\Rightarrow$ $r_1 = 10$ cm

Similarly, area of second region $= 100\pi = \pi\,(r_2^2 - r_1^2)$

$100\pi = \pi\,(r_2^2 - 100)$

$\Rightarrow$ $r_2 = 10\sqrt{2}$ cm

Again, area of third region $= 100\pi = \pi\,(r_3^2 - r_2^2)$

$100\pi = \pi\,(r_3^2 - 200) \Rightarrow r_3^2 = 10\sqrt{3}$cm

$\therefore$ The required radii are 10, $10\sqrt{2}$ and $10\sqrt{3}$.

61 $\angle DAB + \angle DCB = 180°$

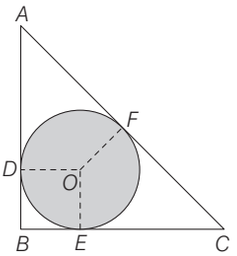
$\angle BAO + \angle OAD + \angle DCO = 180°$

$\angle 69° + (\angle OAD + \angle DCO) = 180°$

$\angle OAD + \angle DCO = 111°$

Hence choice (b) is the correct one.

62 $AC = \sqrt{2}$ cm, so $AB = BC = 1$ cm.



The inradius $OD = OE = OF$

$$= \frac{AB + BC - AC}{2} = \frac{2 - \sqrt{2}}{2} \text{ cm}$$

Therefore $DE = \sqrt{2}\left(\frac{2-\sqrt{2}}{2}\right)$

$= \sqrt{2} - 1$ cm

Hence choice (a) is the correct one.

63 Using Alternate segment theorem, we get

$\angle ABC = \angle QAC = 48°$

$\angle BAP = \angle ACB = 64°$

$\angle ABP = \angle ACB = 64°$

$\therefore \angle APB = 180 - (\angle BAP + \angle ABP) = 52°$

And $BAC = 180 - (\angle ACB + \angle ABC) = 68°$

Opposite angles of a cyclic quadrilateral are always supplementary, therefore

$\angle BDC = 180 - \angle BAC = 112°$

Now, $\angle CBP = \angle ABC + \angle ABP = 48° + 64° = 112°$

Thus, $\angle BDC + \angle CBP + \angle APB$

$= 112° + 112° + 52° = 276°$

Hence choice (b) is the correct answer.

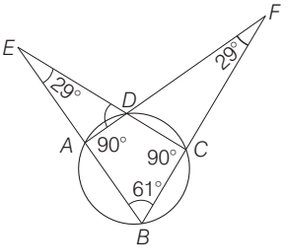
64 $\angle ECB = 180° - (\angle CEB + \angle EBC) = 90°$

But since A, B, C, D are cyclic, so $\angle DAB + \angle DCB$

$= 180° \Rightarrow \angle DAB = 90°$

Again, $\angle AFB = 180° - (\angle FAB + \angle FBA) = 29°$

Since, $\angle EAD = 180° - \angle BAD = 90°$



and $\angle EDA = 180° - (\angle DEA + \angle EAD) = 61°$

$\therefore$ $\alpha - \beta = 61 - 29 = 32°$

Hence choice (c) is the correct one.

Alternatively

$\alpha = \angle ADE = \angle ABC = 61°$

(As $\angle ADE + \angle ADC = \angle ABC + \angle ADC = 180°$)

$\therefore$ $\angle CDF = 61°$

and $\angle ACF = \angle BEC + \angle EBC = 90°$

($\angle ACE$ is an external angle of ΔEBC)

$\therefore \quad \beta = \angle CFD = 180° - (\angle CDF + \angle DCF)$

$= 180° - (61° + 90°) = 29°$

$\therefore \quad \alpha - \beta = 61° - 29° = 32°$

65 **Case I :** $\angle OBC = \angle OCB = 42°$

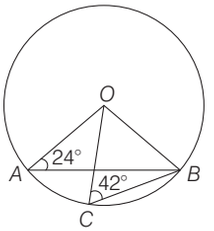
$\therefore \quad \angle BOC = 180 - 2(42°) = 96°$

and $\angle OBA = \angle OAB = 24°$

$\therefore \quad \angle AOB = 180 - 2(24°) = 132°$

Thus, $\angle AOC = \angle AOB - \angle BOC$

$\Rightarrow \quad \angle AOC = 132 - 96° = 36°$



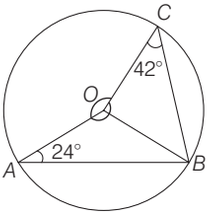
Case II : $\angle AOB = 180 - 2(24) = 132°$

and $\angle BOC = 180° - 2(42) = 96°$

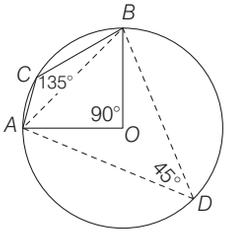
Thus, $\angle AOC = \angle AOB + \angle BOC$

$\Rightarrow \quad \angle AOC = 132 + 96 = 228°$

Hence choice (d) is the correct one.



66 First of all complete the circle as shown below. Draw the chord AB and consider an arbitrary point D lying on the major arc of the circle.



Now, central angle $AOB = 90°$, so the angle in the major segment due to the same chord AB is angle $ADB = 45°$.

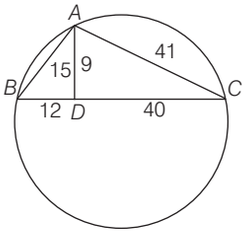
But since the quadrilateral $ADBC$ is a cyclic one, therefore

$\angle ACB + \angle ADB = 180°$

$\Rightarrow \quad \angle ACB + 45° = 180° \Rightarrow \angle ACB = 135°$

Hence choice (b) is the correct one.

67 Let's drop a perpendicular AD on BC, we will have two right angled triangles ΔADB and ΔADC.



By using Pythagoras theorem we can determine AD, BD and CD, as follows

$AD^2 = AB^2 - BD^2$

$= AB^2 - (BC - CD)^2$

$= 15^2 - (52 - CD)^2 \quad \ldots(i)$

Also, $AD^2 = AC^2 - CD^2$

$= 41^2 - CD^2$

Comparing eq. (i) and eq. (ii), we get

$CD = 40$ cm, $BD = 12$ cm and $AD = 9$ cm

Now, the area of triangle $ABC = \frac{1}{2} \times AD \times BC$

$= \frac{1}{2} \times 9 \times 52 = 234$ sq cm

Let us consider that the circumradius of the triangle is R and area is A, then we have

$A = \frac{abc}{4R}$

$\Rightarrow \quad 234 = \frac{15 \times 41 \times 52}{4R}$

$\Rightarrow \quad R = \frac{205}{6}$ cm

Therefore, the diameter of the circumcircle $= \frac{205}{3}$ cm

Hence choice (a) is the correct one.

68 Curvature of a circle

$= \frac{1}{\text{radius of the circle}} = \frac{1}{2} = 0.5$ cm

Hence choice (c) is the correct one.

69 Since the radius of a circle

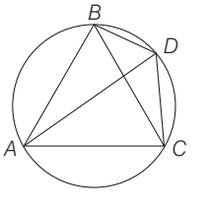
$= \frac{1}{\text{curvature of the circle}}$

Therefore, radii of the circles are $\frac{1}{0.5} = 2$ and $\frac{1}{2} = 0.5$

Thus the required distance between the centers of the circles $= 2 + 0.5 = 2.5$

Hence choice (a) is the correct one.

70 Draw the following diagram and you will find that $ABDC$ is a cyclic quadrilateral. However, the Ptolemy's theorem states that, "If a quadrilateral is inscribable in a circle then the product of the measures of its diagonals is equal to the sum of the products of the measures of the pairs of opposite sides."



Therefore, $AB \cdot CD + AC \cdot BD = BC \cdot AD$

$\Rightarrow \quad 30(CD) + 30(BD) = 30(AD)$

$\Rightarrow \quad CD + BD = AD$

$\Rightarrow \quad CD + BD = 32$ cm

Hence choice (b) is the correct one.